# JOSHUA CARBON

**COMPUTER ENGINEER / AI DEVELOPER**

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## SUMMARY

Computer Engineering graduate with practical experience in Artificial Intelligence, Computer Vision, Software Development, and Embedded Systems through academic, internship, capstone, and personal projects. Built AI-powered solutions using Python, YOLOv5, OpenCV, Roboflow, Google Colab, and Microsoft Copilot Studio, alongside embedded and IoT systems. Comfortable with debugging, system integration, and technical documentation.

## EXPERIENCE

**Intern/OJT — Erovoutika Robotics and Automation Solution** | *May 2023 – Aug 2023*

- Collaborated with engineering teams to troubleshoot technical issues across automation and embedded systems, contributing to improved system efficiency.
- Performed debugging and calibration for automation initiatives, ensuring stable operation of integrated hardware and software components.
- Supported integration of sensors, embedded components, and control systems, building a foundation later applied to AI and IoT project work.
- Authored technical documentation for project guidelines, supporting future engineering and development reference.

## SKILLS

**Programming Languages:** Python, C++, C, C#
**Artificial Intelligence & Computer Vision:** YOLOv5, OpenCV, OCR, Image Processing, Roboflow, Google Colab
**AI Tools:** Microsoft Copilot Studio, Prompt Engineering, Power Automate, OpenAI API
**Software Development:** Object-Oriented Programming, Windows Forms, Microsoft Visual Studio, Visual Studio Code
**Embedded Systems & IoT:** Raspberry Pi, Arduino, ESP32, Blynk, Sensor Integration, Embedded C, PCB Design
**Other:** Git, MATLAB, Simulink, Proteus, Tinkercad, Fritzing

## PROJECTS

**AI-Based Japanese Character Recognition using YOLOv5 (Thesis Project) —** Developed an AI-powered Optical Character Recognition (OCR) system using a custom-trained YOLOv5 model to recognize Japanese Kanji characters. The project reported 95.33% accuracy across 50 Kanji classes and 150 test trials.

**CompetencyIQ AI —** Created an AI-powered workforce intelligence assistant using Microsoft Copilot Studio designed to analyze employee competency, training, certifications, compliance, findings, and operational risk.

**QA Automation Framework —** Built a Python, Robot Framework, and Selenium automation framework with reusable components, automated reporting, and optional OpenAI API-based test-result analysis.

**Smart Parking System —** Developed an IoT-based parking monitoring system using sensors and an ESP8266 controller to detect parking occupancy and display real-time slot availability through a Blynk mobile application.

## EDUCATION

Bachelor of Science, Computer Engineering — Mapúa University | 2019 – 2025
Bachelor of Science, Electrical Engineering | 2014 – 2019

## ADDITIONAL

- Familiar with Software Development Life Cycle (SDLC) and willing to learn.
- Familiar with Object-Oriented Programming (OOP) concepts and eager to deepen understanding.
- Interested in Artificial Intelligence, Machine Learning, Automation, and related engineering technologies.